Notes: Grosfeld, Sakalli & Zhuravskaya (RES, 2020)

Middleman Minorities and Ethnic Violence: Anti-Jewish Pogroms in the Russian Empire

Contents

1 Big picture

- **Question.** Why are middleman minorities – minorities specializing in trade, finance, and market intermediation – sometimes targeted by ethnic violence?
- **Setting.** The paper studies anti-Jewish pogroms in the Pale of Settlement in the Russian Empire from 1800 to 1927.
- **Historical feature.** Jews were a small minority in local population but were disproportionately represented in middleman occupations, especially moneylending and grain trade.
- **Main finding.** Pogroms occurred when economic shocks coincided with political turmoil, and especially in localities where Jews dominated agriculturally related middleman occupations.
- **Two relevant economic shocks.** Local shocks are measured by local crop failure / extreme spring heat. Marketwide shocks are measured by severe crop failures in major grain-producing regions and by grain-price shocks.
- **Political turmoil.** Turmoil captures moments of extreme uncertainty about political order: the assassination of Tsar Alexander II, the Russo-Japanese War and 1905 revolution, World War I, the revolutions of 1917, and the Civil War.
- **Rejected simple explanation.** The evidence is not consistent with a pure scapegoating hypothesis in which Jews were blamed for all misfortunes. If scapegoating alone were central, the general Jewish share or all Jewish occupations should matter similarly.
- **Proposed mechanism.** Jewish middlemen provided insurance and intermediation to peasants and grain buyers in repeated relationships. In stable times, future cooperation made debt rollovers and informal insurance sustainable. In political turmoil, commitment collapsed, creditors and traders demanded immediate repayment, and violence broke out.
- **Occupation-specific result.** Violence is tied to Jewish dominance among creditors and grain traders, not to Jewish dominance in unrelated occupations such as craft, transport, or non-agricultural trade.
- **Key empirical design.** The paper uses grid-cell-by-year panel data with grid and year fixed effects, climate and grain-price shocks, census occupational shares, and robust spatial-temporal standard errors.
- **Takeaway.** Ethnic violence against middleman minorities can arise not because middlemen directly compete with the majority, but because a breakdown in repeated economic relationships converts complementary intermediation into conflict.

2 Data and measurement

2.1 Geography: the Pale of Settlement

- The Pale of Settlement was the large region of the Russian Empire where Jews were legally permitted to reside.
- It covered parts of present-day Eastern Europe, including areas in modern Ukraine, Belarus, Lithuania, Latvia, Poland, Moldova, and Russia.
- The paper divides the territory into 0.5×0.5 degree grid cells.
- The main cross-sectional unit is a grid cell, and the main panel unit is grid cell $\times$ year.
- The baseline panel contains 576 grid cells and 73,728 grid-cell-year observations.

Interpretation: The grid-cell design avoids relying only on administrative borders, which changed over time, and allows the authors to combine historical pogrom data, weather data, grain shocks, and census occupation data at a common spatial scale.

2.2 Pogrom data

- The outcome is whether at least one anti-Jewish pogrom occurred in a grid cell in a given year.
- Pogroms are mob violence against Jews; the data are built from historical records and secondary sources.
- The first major pogrom in the data was in Odessa in 1821.
- Most pogroms occurred in three waves: 1881–1882, 1903–1906, and 1917–1921.
- The mean probability of a pogrom in any grid cell-year is about 0.52%.

Interpretation: Pogroms are rare events, so the paper multiplies the dependent variable by 100 and interprets coefficients as percentage-point effects on pogrom probability.

2.3 Jewish population and occupation shares

- The key cross-sectional data come from the 1897 Russian Empire census.
- The census reports local Jewish population shares and the shares of Jews among workers in different occupations.
- The paper focuses especially on Jews among creditors and Jews among grain traders.
- Other occupations include non-agricultural traders, general traders, craftsmen, transporters, and peasants.
- The average Jewish share in local population is about 10.3%, but Jewish shares in several middleman occupations are much higher.

Interpretation: The distinction between the Jewish population share and Jewish occupational shares is crucial. The paper asks whether violence was connected to Jews as Jews, or specifically to Jews serving as middlemen in agriculture-related credit and trade.

2.4 Local economic shocks

- A local economic shock is designed to proxy for local crop failure.
- The main measure uses climate data: extreme deviations of spring temperature from the grid-cell historical mean.
- Extremely hot springs are associated with grain harvest failures and hence stress for agricultural producers.
- The paper also checks alternative local-shock measures, including government grain-assistance data and reconstructed climate data.

Interpretation: Local shocks matter for the creditor channel because peasants facing poor harvests may be unable to repay loans. The middleman relationship becomes fragile when bad harvests coincide with political uncertainty.

2.5 Marketwide economic shocks

- A marketwide economic shock captures shocks to the broader grain market.
- The main measure uses severe crop failures in major grain-producing regions.

- The paper also uses grain prices, especially rye prices, as an alternative marketwide shock measure.
- Marketwide shocks are relevant for grain traders because local buyers and sellers face price spikes and grain-market disruption.

Interpretation: Marketwide shocks matter for the grain-trader channel. Jewish grain traders were intermediaries between producers, consumers, and markets; price spikes and shortages strained these relationships.

2.6 Political turmoil

- Political turmoil is a dummy for periods of severe uncertainty about the future political and social order.
- The key episodes include the assassination of Tsar Alexander II, the Russo-Japanese War and the 1905 revolution, the First World War, the 1917 revolutions, and the Civil War.
- Political turmoil alone is not enough: the main mechanism requires turmoil interacting with economic shocks.

Interpretation: Political turmoil reduces the value of future relationships and makes enforcement of informal contracts weaker. This is why economic shocks during turmoil can produce violence even when similar shocks in stable times do not.

2.7 Pre-existing antisemitism and controls

- The paper controls for measures of pre-existing antisemitism, such as past anti-Jewish violence and the local presence of antisemitic books in the 18th century.
- It also uses occupational segregation indexes and controls for the general Jewish population share.
- Robustness checks include spatial matching, alternative shock definitions, alternative political turmoil measures, subsamples, alternative aggregation levels, and placebo outcomes.

Interpretation: These controls address the concern that places with Jewish middlemen were simply more antisemitic or more segregated. The main results remain after accounting for these pre-existing conditions.

3 Empirical specifications

3.1 Baseline shock interaction model

The baseline panel regression is:

$$V_{it} = \alpha_i + \mu_t + \beta E_{it} + \gamma P_t + \delta(E_{it} \times P_t) + \varepsilon_{it}, \tag{1}$$

where:

- V_{it} is pogrom occurrence in grid cell i and year t, multiplied by 100;
- E_{it} is a local or marketwide economic shock;
- P_t is political turmoil;
- α_i are grid-cell fixed effects;
- μ_t are year fixed effects.

Interpretation: The coefficient δ tests whether economic shocks become violent specifically during political turmoil. The main result is that shocks matter much more when political order is uncertain.

3.2 Middleman-minority triple interactions

The paper then estimates specifications of the form:

$$V_{it} = \alpha_i + \mu_t + \delta(E_{it} \times P_t \times M_i) + \Gamma' Z_{it} + \varepsilon_{it}, \tag{2}$$

where M_i is an occupation-specific Jewish middleman share, such as the share of Jews among creditors or grain traders.

Interpretation: The coefficient on the triple interaction asks whether shocks during turmoil are more likely to produce pogroms in locations where Jews dominated the relevant middleman occupation.

3.3 Creditor channel

For local harvest shocks, the relevant middleman occupation is moneylending:

$$V_{it} = \alpha_i + \mu_t + \delta_c(LocalShock_{it} \times PoliticalTurmoil_t \times JewishCreditors_i) + \cdots . \tag{3}$$

Interpretation: A positive δ_c means local crop failures during turmoil are more likely to produce pogroms where Jews were important creditors. This supports the repeated-credit-relationship mechanism.

3.4 Grain-trader channel

For marketwide shocks, the relevant occupation is grain trade:

$$V_{it} = \alpha_i + \mu_t + \delta_g(MarketwideShock_t \times PoliticalTurmoil_t \times JewishGrainTraders_i) + \cdots . \tag{4}$$

Interpretation: A positive δ_g means marketwide grain shocks during turmoil are more likely to produce pogroms where Jews dominated grain trading. This supports the agricultural market-intermediation mechanism.

3.5 Scapegoating test

The scapegoating hypothesis predicts that violence should be related to the general share of Jews:

$$V_{it} = \alpha_i + \mu_t + \delta_s(E_{it} \times P_t \times JewishShare_i) + \cdots . \tag{5}$$

Interpretation: The paper finds that the general Jewish population share does not explain the main effects once occupation-specific Jewish middleman shares are included. This weakens the pure scapegoating interpretation.

3.6 Spatial matching specification

The spatial matching analysis compares neighboring grid cells that are similar geographically but differ in Jewish middleman shares. The treatment is high Jewish participation in creditors or grain traders, and controls are neighboring grid cells with lower participation.

Interpretation: Matching addresses the concern that Jewish middlemen were concentrated in systematically different places. The results remain positive in geographically local comparisons.

4 Identification and empirical design

4.1 Sources of variation

- **Time variation:** political turmoil and marketwide economic shocks vary across years.
- **Space-time variation:** local crop shocks vary across grid cells and years.
- **Cross-sectional variation:** Jewish occupational shares vary across grid cells.
- **Fixed effects:** grid fixed effects absorb time-invariant local differences; year fixed effects absorb shocks common to all grid cells.

Interpretation: The key identifying variation is whether locations with higher Jewish middleman shares are differentially affected by economic shocks during political turmoil.

4.2 Why the design separates mechanisms

- The creditor channel predicts effects for local shocks and Jewish moneylenders.
- The grain-trader channel predicts effects for marketwide grain shocks and Jewish grain traders.
- The scapegoating hypothesis predicts effects of the general Jewish share across many shocks and occupations.
- The economic-competition hypothesis predicts violence when Jews competed with the majority, but the occupations studied here were complementary to the majority's activities.

Interpretation: The paper's occupation-specific patterns are the main evidence for the middleman-relationship mechanism.

4.3 Inference

- Standard errors are corrected for spatial and temporal correlation following Hsiang's procedure.
- The main regressions use a spatial radius and a temporal lag to allow nearby observations and adjacent years to be correlated.
- Robustness checks also use alternative clustering and sample definitions.

Interpretation: Since pogroms and shocks are spatially and temporally clustered, standard independent errors would understate uncertainty.

4.4 Limitations

- Occupation shares are measured from the 1897 census, while pogroms span 1800–1927.
- Some measures of antisemitism and economic conditions are proxies rather than direct observations.
- Pogrom recording may be incomplete, especially for earlier periods.
- The paper identifies conditions associated with pogrom outbreaks, not every micro-level decision that led individuals to participate in violence.

Interpretation: The paper's strength is that many different pieces of evidence point to the same occupation-specific mechanism.

5 Tables and figures

5.1 Figure 1: Geography of pogroms in the Pale of Settlement

Read:

- The map shows the Pale of Settlement and the geographic distribution of anti-Jewish pogroms.
- Green circles indicate locations of pogroms, with circle size proportional to the number of pogroms.
- Pogroms are geographically widespread but concentrated in parts of present-day Ukraine, Belarus, Lithuania, and surrounding regions.

Economic message: The map motivates the paper's spatial design: the authors need to explain why violence occurred in some places and not others within the same broad institutional environment.

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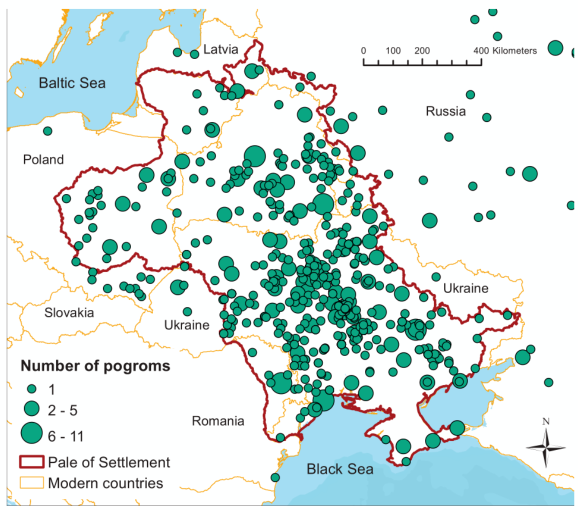

FIGURE 1

The Pale of Settlement and the geographic distribution of pogroms.

tes: This map presents the geographic distribution of pogroms in Eastern Europe. The darker thicker line represents the border of

5.2 Figure 2 and Figure 3: timing of shocks and middleman exposure

Read:

- Figure 2 shows that pogrom waves align with moments when economic shocks coincided with political turmoil.
- Crop failures alone did not mechanically cause pogroms; some severe agricultural shocks occurred in politically stable periods without large pogrom waves.
- Figure 3 shows that pogrom frequency was higher in locations where Jews had larger shares among creditors and grain traders.
- The pattern is strongest for grain traders during years with both political turmoil and marketwide economic shocks.

Economic message: Economic distress had to interact with political instability and relevant Jewish middleman positions. This is the core evidence against a simple one-factor explanation.

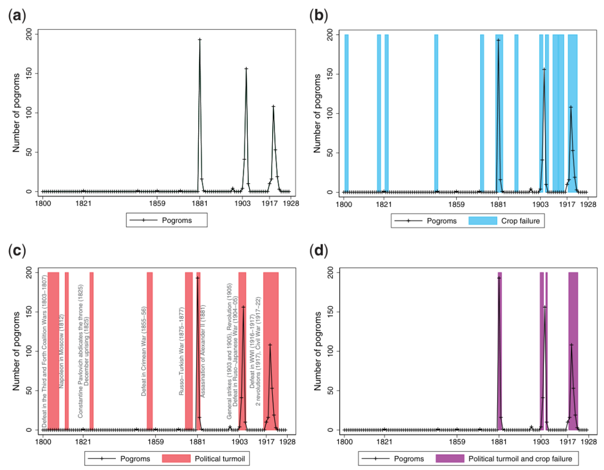

FIGURE 2

Pogrom waves, marketwide economic shocks, and political turmoil. (a) Pogrom occurrence; (b) Pogrom occurrence and marketwide economic shocks; (c) Pogrom occurrence and political turmoi; (d) Pogrom occurrence at the intersection of

Figure 2: Pogrom waves, economic shocks, and political turmoil

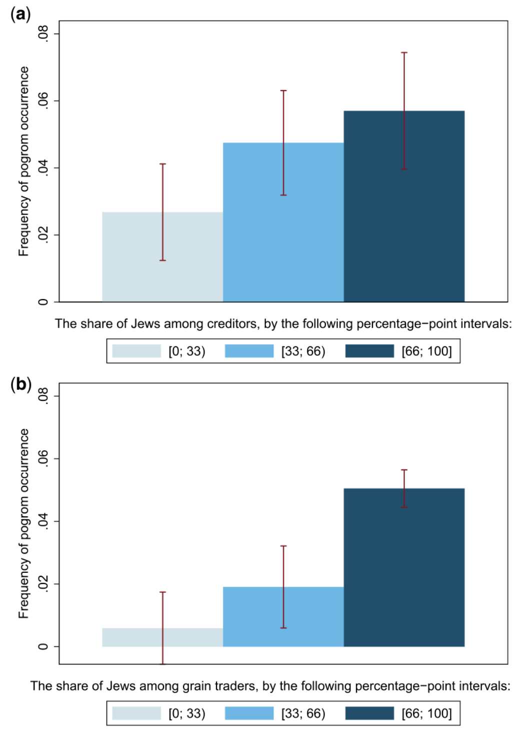

FIGURE 3

Figure 3: Pogroms and Jewish middlemen shares

5.3 Table 1 and Table 2: Summary statistics and shock construction

Read:

- Table 1 shows that Jews were 10.3% of the local population on average but had very high shares in trader and creditor occupations.
- Jews accounted for about 53.4% of creditors and 88.2% of grain traders on average.
- Pogrom occurrence in a grid cell-year averaged 0.52%.
- Table 2 verifies that local economic shock measures are related to grain harvests and that marketwide shocks are related to grain prices.

Economic message: Table 1 establishes the middleman-minority setting, and Table 2 validates the economic-shock measures used in the main regressions.

TABLE 1

Summary statistics: pogroms and Jewish presence in different occupations

Variable	Mean	Std. Dev.	Min.	Max.
Panel A: The share of Jews in local population and in occupations across grid cells				
		Number of observations = 576		
Share of Jews in population	0.103	0.051	0.009	0.249
Share of Jews among creditors	0.534	0.281	0	1
Share of Jews among traders	0.800	0.209	0.064	0.988
Share of Jews among agricultural traders	0.812	0.207	0.046	0.995
Share of Jews among grain traders	0.882	0.175	0.176	1
Share of Jews among non-agricultural traders	0.820	0.198	0.087	0.993
Share of Jews among general traders	0.791	0.226	0.068	0.995
Share of Jews among craftsmen	0.481	0.208	0.035	0.841
Share of Jews among transporters	0.350	0.209	0.003	0.913
Share of Jews among peasants	0.007	0.006	0	0.041
Panel B: Pogroms across grid cell × year observations				
		Number of observations = 73,728		
Pogrom occurrence	0.0052	0.0721	0	1
Pogrom occurrence in agricultural season	0.0035	0.0591	0	1
Pogrom occurrence in harvest period	0.0017	0.0408	0	1
Pogrom occurrence in non-agricultural season	0.0015	0.0386	0	1
Pogrom occurrence in unknown season	0.0006	0.0241	0	1
Number of pogroms	0.0087	0.1681	0	20
Number of pogroms in agricultural season	0.0059	0.1455	0	19
Number of pogroms in harvest period	0.0024	0.0689	0	7
Number of pogroms in non-agricultural season	0.0020	0.0555	0	4
Number of pogroms in unknown season	0.0007	0.0317	0	3

Table 1: Summary statistics

TABLE 2

Grain yield, grain price, and the local and marketwide economic shocks

	Log grain harvest: 1864–1914			Log price of rye: 1860–1915				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local econ shock	−0.209***	−0.213***	−0.196**	0.012	−0.009			
	(0.075)	(0.070)	(0.080)	(0.017)	(0.021)			
Local econ shock lag		0.028		0.016	−0.015			
		(0.114)		(0.019)	(0.020)			
Local econ shock × Political turmoil			−0.033					
			(0.170)					
Local econ shock × High grain suitability					0.035*			
					(0.021)			
Local econ shock lag × High grain suitability					0.061*			
					(0.031)			
Marketwide econ shock						0.053**	0.063***	0.077***
						(0.023)	(0.023)	(0.027)
Political turmoil							−0.055**	−0.026
							(0.022)	(0.027)
Marketwide econ shock × Political turmoil								−0.066
								(0.044)
High grain suitability					−0.039***			
					(0.008)			
Log grain price lag				0.745***	0.695***			
				(0.029)	(0.032)			
R^2	0.593	0.593	0.593	0.804	0.810	0.013	0.022	0.026
Year FE	Yes	Yes	Yes	Yes	Yes	No	No	No
Province FE	Yes	Yes	Yes	No	No	No	No	No
Observations	535	535	535	1,294	1,294	1,312	1,312	1,312
Mean of dependent var.	15.47	15.47	15.47	3.865	3.865	3.866	3.866	3.866
SD of dependent var.	0.724	0.724	0.724	0.216	0.216	0.216	0.216	0.216

Table 2: Grain yield, grain price, and economic shocks

5.4 Table 3 and Figure 4: Shocks, turmoil, and pogrom occurrence

Read:

- Table 3 shows that local and marketwide economic shocks are associated with pogrom occurrence, but the interaction with political turmoil is central.
- The local shock × political turmoil coefficient is large and positive.
- The marketwide shock × political turmoil coefficient is also large and positive.
- Figure 4 shows a sharp increase in pogrom probability at extreme hot spring temperatures, especially during political turmoil.

Economic message: Economic shocks become violent when they occur during periods of political instability. This supports the idea that uncertainty about future enforcement and relationships is crucial.

TABLE 3

Pogrom occurrence at the intersection of economic shocks and political turmoil

	Pogrom occurrence						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Local econ shock	1.757***		0.003		1.088*	0.047	0.058
	(0.474)		(0.038)		(0.584)	(0.061)	(0.062)
Marketwide econ shock		1.374***		0.002			
		(0.205)		(0.022)			
Political turmoil			1.799***	0.861***			
			(0.214)	(0.162)			
Local econ shock × Political turmoil			2.752***			2.937*	3.274**
			(0.971)			(1.620)	(1.602)
Marketwide econ shock × Political turmoil				3.810***			
				(0.591)			
Local econ shock × Political turmoil × Share of Jews							33.637*
							(19.367)
Political turmoil × Share of Jews							0.335
							(1.610)
Local econ shock × Share of Jews							−0.105
							(0.859)
Marketwide econ shock × Political turmoil × Share of Jews							−0.856
							(9.646)
Marketwide econ shock × Share of Jews							−0.049
							(0.592)
R^2	0.016	0.019	0.032	0.043	0.113	0.114	0.115
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	Yes	Yes	Yes
Observations	73,728	73,728	73,728	73,728	73,728	73,728	73,728
Mean of dependent var.	0.522	0.522	0.522	0.522	0.522	0.522	0.522
SD of dependent var.	7.207	7.207	7.207	7.207	7.207	7.207	7.207

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. This table presents the results of regressions in which the probability of a pogrom in a grid cell and year is related to local economic shocks, marketwide economic shocks, and political turmoil. Standard errors are corrected for both spatial and temporal correlations following Hsiang (2010) in a radius of 100 km and 1 temporal lag. ***$P<0.01$. **$P<0.05$. *$P<0.1$.

Table 3: Pogroms at the intersection of shocks and turmoil

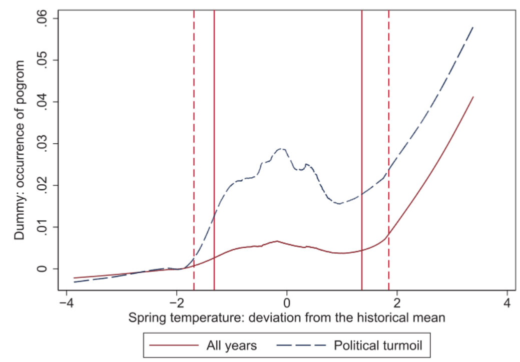

FIGURE 4

Pogroms, spring temperature, and political turmoil.

re presents unconditional nonparametric locally weighted regressions (LOWESS) between pogrom occu
d the deviation of spring temperature in a grid cell and year from the historical mean (for all years and fo
l). From left to right, the dashed vertical lines indicate the 5th and the 95th percentiles of the distribution

Figure 4: Spring temperature and pogroms

5.5 Table 4: Jewish dominance in moneylending and grain trade

Read:

- The triple interaction of local shock, political turmoil, and Jewish share among creditors is positive and statistically significant.
- The triple interaction of marketwide shock, political turmoil, and Jewish share among grain traders is also positive and statistically significant.
- The general Jewish population share does not explain the main result once middleman shares are included.

Economic message: The evidence points to occupation-specific economic relationships. Violence was more likely where Jews were crucial agriculturally related middlemen, not merely where there were more Jews.

TABLE 4

Pogroms and the Jewish domination over moneylending and grain trade

	Pogrom occurrence				
	(1)	(2)	(3)	(4)	(5)
Local econ shock × Political turmoil × Sh. Jews among creditors	6.56***				5.95**
	(2.47)				(2.48)
Local econ shock × Political turmoil × Sh. Jews among grain traders		0.82			
		(4.41)			
Marketwide econ shock × Political turmoil × Sh. Jews among grain traders			9.38***	9.42***	10.13***
			(2.75)	(2.69)	(2.83)
Political turmoil × Sh. Jews among creditors	0.34				0.08
	(0.62)				(0.61)
Local econ shock × Sh. Jews among creditors	−0.01				−0.02
	(0.10)				(0.10)
Political turmoil × Sh. Jews among grain traders		3.03***	0.05		
		(0.91)	(0.56)		
Local econ shock × Sh. Jews among grain traders		0.21			
		(0.19)			
Marketwide econ shock × Sh. Jews among grain traders			0.00		
			(0.15)		
Local econ shock × Political turmoil	3.34**	3.32**			4.23***
	(1.59)	(1.57)			(1.59)
Local econ shock × Political turmoil × Share of Jews	20.51	38.02*			26.53
	(18.05)	(21.00)			(19.61)
R^2	0.118	0.119	0.118	0.118	0.123
Grid and year FE	Yes	Yes	Yes	Yes	Yes
Other interactions with the share of Jews	Yes	Yes	Yes	Yes	Yes
Interactions with the share of creditors in total employed	Yes	No	No	No	Yes
Interactions with the share of grain traders in total employed	No	Yes	Yes	Yes	Yes
Observations	73,728	73,728	73,728	73,728	73,728
Mean of dependent var.	0.522	0.522	0.522	0.522	0.522
SD of dependent var.	7.207	7.207	7.207	7.207	7.207

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. This table presents OLS regression results in which the probability of a pogrom in a grid cell and year is related to the local shares of Jews among moneylenders and grain traders, local economic shocks, marketwide economic shocks, and political turmoil, controlling for year and grid-cell fixed effects as well as the share of Jews and the shares of middleman sectors in local employment interacted with economic and political shocks. Standard errors are corrected for both spatial and temporal correlations following Hsiang (2010) in a radius of 100 km and 1 temporal lag. ***$P<0.01$. **$P<0.05$. *$P<0.1$.

5.6 Table 5: Comparing all Jewish occupations

Read:

- Panel A compares local shocks across Jewish occupational shares.
- The creditor effect remains positive and significant when other occupations are considered.
- Panel B compares marketwide shocks across occupations.
- The grain-trader effect is the robust positive effect for marketwide economic shocks.
- Other occupations do not show a similar stable positive pattern; some coefficients are negative or insignificant.

Economic message: The paper's mechanism is specific to middleman occupations related to agriculture. It is not a generic result about Jewish employment or Jewish economic prominence.

TABLE 5

The shares of Jews in all major occupations, economic shocks, and political turmoil

Panel A: The effects of local economic shocks	Pogrom occurrence							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local econ shock × Pol. turmoil × Sh. Jews among creditors	6.24^{***}							6.92^{**}
	(2.36)							(2.87)
Pol. turmoil × Sh. Jews among creditors	−0.73							−0.07
	(0.55)							(0.64)
Local econ shock × Pol. turmoil × Sh. Jews among grain traders		2.18						−0.88
		(4.45)						(5.53)
Pol. turmoil × Sh. Jews among grain traders		2.35^{***}						3.79^{***}
		(0.86)						(0.98)
Local econ shock × Pol. turmoil × Sh. Jews among non-agric. traders			2.29					−6.38
			(3.55)					(5.59)
Pol. turmoil × Sh. Jews among non-agric. traders			0.29					0.56
			(0.92)					(1.59)
Local econ shock × Pol. turmoil × Sh. Jews among general traders				3.18				5.70
				(3.47)				(6.72)
Pol. turmoil × Sh. Jews among general traders				0.10				−0.48
				(0.77)				(1.45)
Local econ shock × Pol. turmoil × Sh. Jews among craftsmen					8.50^{*}			1.54
					(5.13)			(7.31)
Pol. turmoil × Sh. Jews among craftsmen					-3.83^{***}			-4.06^{***}
					(1.25)			(1.58)
Local econ shock × Pol. turmoil × Sh. Jews among transporters						1.83		−3.75
						(3.29)		(3.87)
Pol. turmoil × Sh. Jews among transporters						−1.30		−0.97
						(0.83)		(1.05)
Local econ shock × Pol. turmoil × Sh. Jews among peasants							131.37	122.65
							(159.61)	(156.28)
Pol. turmoil × Sh. Jews among peasants							-78.33^{***}	-44.05^{**}
							(23.74)	(21.60)
R^2	0.117	0.116	0.116	0.116	0.116	0.116	0.116	0.119

Table 5, Panel A: Local shocks and occupations

TABLE 5

(Continued)

Panel B: The effects of marketwide economic shocks	Pogrom occurrence							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Marketwide econ shock × Pol. turmoil × Sh. Jews among creditors	1.52							2.14
	(1.48)							(1.83)
Marketwide econ shock × Pol. turmoil × Sh. Jews among grain traders		8.19^{***}						9.41^{***}
		(2.51)						(2.89)
Marketwide econ shock × Pol. turmoil × Sh. Jews among non-agric. traders			4.01					2.09
			(2.47)					(3.91)
Marketwide econ shock × Pol. turmoil × Sh. Jews among general traders				2.86				−1.43
				(2.08)				(3.66)
Marketwide econ shock × Pol. turmoil × Sh. Jews among craftsmen					−2.66			−6.32
					(3.40)			(4.57)
Marketwide econ shock × Pol. turmoil × Sh. Jews among transporters						−0.50		−2.86
						(2.29)		(2.97)
Marketwide econ shock × Pol. turmoil × Sh. Jews among peasants							-134.90^{*}	−71.28
							(75.10)	(69.55)
R^2	0.114	0.116	0.114	0.114	0.114	0.113	0.114	0.117
Grid and year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interactions with the share of Jews	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interactions with local economic shocks in Panel A	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	73,728	73,728	73,728	73,728	73,728	73,728	73,728	73,728
Mean of dependent var.	0.522	0.522	0.522	0.522	0.522	0.522	0.522	0.522
SD of dependent var.	7.207	7.207	7.207	7.207	7.207	7.207	7.207	7.207

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. This table presents OLS regression results in which the probability of a pogrom in a grid cell and year is related to the economic shocks, political turmoil, and the shares of Jews among the locally employed in different occupations, controlling for year and grid-cell fixed effects, and the interactions of the share of Jews with economic and political shocks. Panel A focuses on local economic shocks and Panel B focuses on marketwide economic shocks. Standard errors are corrected for both spatial and temporal correlations

Table 5, Panel B: Marketwide shocks and occupations

5.7 Figure 5 and Table 6: Reaction of Jews and pre-existing antisemitism

Read:

- Figure 5 shows that places with more pogroms had subsequent declines in the share of Jews, including the share of Jews among middlemen.
- Table 6 shows correlations between Jewish occupational structure and measures of past antisemitism.
- Some middleman measures are correlated with past violence or antisemitic books, but these correlations do not explain the main regressions once controls are included.

Economic message: Pogroms had consequences for Jewish presence and occupational structure. At the same time, the main mechanism is not reducible to pre-existing antisemitism.

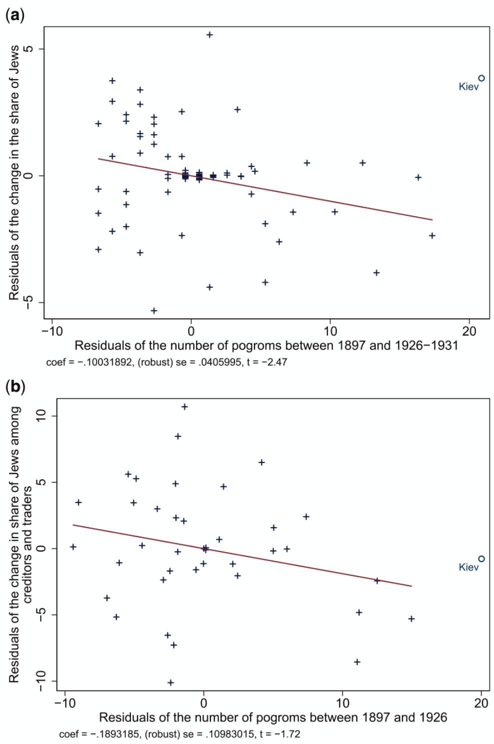

FIGURE 5

:tion of Jews to the pogroms. (a) Correlation between the number of pogroms and the change in the share of Jews
l Pale). (b) Correlation between the number of pogroms and the change in the share of Jews among middlemen.

This figure presents the partial correlation between the number of pogroms, on the one hand, and the change in the share of Jews
A) or the change in the share of Jews among middlemen (Panel B), on the other hand, from the OLS estimates. The scatter plot
el A is conditional on country fixed effects; and that in Panel B is conditional on country fixed effects, the change in the share of

Figure 5: Reaction of Jews to pogroms

TABLE 6

Correlation between past anti-Jewish violence, antisemitic books in 18th century, and Jewish occupations

	Occurrence of past anti-Jewish violence				The share of antisemitic books			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Share of Jews among creditors	9.11	−7.35	−7.89	−0.54	0.011	−0.005	−0.008	−0.010
	(13.79)	(15.71)	(15.97)	(14.28)	(0.017)	(0.018)	(0.018)	(0.019)
Share of Jews among grain traders	44.002***	15.63	22.23	−5.50	0.028*	−0.003	0.000	−0.006
	(15.03)	(16.46)	(14.49)	(15.52)	(0.017)	(0.016)	(0.014)	(0.010)
Occupational segregation index		78.93**				0.082**		
		(36.50)				(0.039)		
Occupational segregation index (w/o middlemen)			75.07**				0.087**	
			(33.17)				(0.036)	
Share of Jews among peasants				−1,702.29***				−0.693
				(587.41)				(0.537)
Share of Jews among non-agr. traders				23.73				−0.013
				(22.34)				(0.023)
Share of Jews among general traders				40.29*				0.042**
				(24.01)				(0.021)
Share of Jews among craftsmen				−49.80**				0.056
				(25.02)				(0.040)
Share of Jews among transporters				49.73***				0.013
				(17.50)				(0.035)
Share of Jews	82.10	67.15	56.91	241.08***	0.233***	0.193***	0.170**	0.086
	(63.23)	(55.64)	(55.50)	(91.81)	(0.073)	(0.067)	(0.068)	(0.112)
Distance to the origin of 1648 uprising (km)	−2.30**	−3.02***	−3.19***	−4.35***				
	(1.05)	(1.10)	(1.13)	(1.02)				
R^2	0.091	0.113	0.113	0.245	0.128	0.146	0.152	0.164
Observations	576	576	576	576	576	576	576	576

Notes: The unit of analysis is a grid cell. In columns 1–4, the dependent variable is a dummy variable that takes the value of 1 if violence against Jews occurred in the grid cell before 1800, and 0 otherwise. We multiply this dependent variable by 100 to measure all coefficients in percentage points of probability of occurrence of past violence against Jews. In columns

Table 6: Antisemitism and occupations

5.8 Table 7 and Table 8: Controls and spatial matching

Read:

- Table 7 adds controls for past anti-Jewish violence, antisemitic books, and occupational segregation.
- The creditor and grain-trader interaction effects remain positive and significant.
- Table 8 uses spatial matching of neighboring grid cells.
- The estimates remain positive in matched local comparisons.

Economic message: The results are not driven by broad occupational segregation, historical antisemitism, or coarse geographic differences.

TABLE 7

Controlling for the measures of preexisting antisemitism and occupational segregation

Panel A: The effect of the local economic shocks	Pogrom occurrence					
	(1)	(2)	(3)	(4)	(5)	(6)
Local econ shock × Pol. turmoil × Sh. Jews among creditors	6.56***	6.89***	5.72**	6.14**	6.18***	5.88**
	(2.47)	(2.51)	(2.46)	(2.51)	(2.36)	(2.31)
Political turmoil × × Sh. Jews among creditors	0.34	−0.01	0.91	0.43	−0.95	−0.77
	(0.62)	(0.60)	(0.65)	(0.61)	(0.71)	(0.69)
Local econ shock × Pol. turmoil × Past violence against Jews		−1.70		−1.48	−1.47	−1.51
		(2.14)		(2.33)	(2.35)	(2.35)
Political turmoil × Past violence against Jews		2.04***		1.96***	1.67***	1.75***
		(0.57)		(0.59)	(0.57)	(0.57)
Local econ shock × Pol. turmoil × Log literate non-Jews × Antisemitic books			0.16	0.14	0.13	0.14
			(0.13)	(0.14)	(0.15)	(0.15)
Political turmoil × Log literate non-Jews × Antisemitic books			0.06*	0.09**	0.12***	0.12***
			(0.04)	(0.04)	(0.04)	(0.04)
Local econ shock × Pol. turmoil × Occupational segregation					−0.15	
					(9.78)	
Political turmoil × Occupational segregation					8.54***	
					(2.23)	
Local econ shock × Pol. turmoil × Occupational segregation (w/o middlemen)						1.72
						(9.45)
Political turmoil × Occupational segregation (w/o middlemen)						7.36***
						(2.10)
R^2	0.118	0.120	0.120	0.122	0.123	0.123
Oster's δ for $\beta = 0$		1.918	1.067	2.239	2.238	1.834

Table 7, Panel A: Controls for local-shock specification

TABLE 7

(Continued)

Panel B: The effect of the marketwide economic shocks	Pogrom occurrence					
	(1)	(2)	(3)	(4)	(5)	(6)
Marketwide econ shock × Pol. turmoil × Sh. Jews among grain traders	9.42***	7.85***	9.36***	7.85***	7.07**	7.41***
	(2.69)	(2.59)	(2.70)	(2.63)	(3.02)	(2.75)
Marketwide econ shock × Pol. turmoil × Past violence against Jews		2.90*		3.40**	3.34**	3.36**
		(1.55)		(1.63)	(1.68)	(1.68)
Marketwide econ shock × Pol. turmoil × Log literate non-Jews × Antisemitic books			0.26***	0.31***	0.32***	0.32***
			(0.08)	(0.08)	(0.09)	(0.09)
Marketwide econ shock × Pol. turmoil × Occupational segregation					1.62	
					(5.60)	
Marketwide econ shock × Pol. turmoil × Occupational segregation (w/o middlemen)						1.05
						(5.12)
R^2	0.118	0.120	0.120	0.122	0.122	0.122
Oster's δ for $\beta = 0$		5.490	−10.05	9.531	1.789	2.442
Grid and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Interactions with the share of Jews and the middleman sector shares	Yes	Yes	Yes	Yes	Yes	Yes
Interactions with total number of books published	No	No	Yes	Yes	Yes	Yes
All lower-level interactions	Yes	Yes	Yes	Yes	Yes	Yes
Observations	73,728	73,728	73,728	73,728	73,728	73,728
Mean of dependent var.	0.522	0.522	0.522	0.522	0.522	0.522
SD of dependent var.	7.207	7.207	7.207	7.207	7.207	7.207

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. The table presents OLS regression results that show the relationship between pogroms and the share of Jews in middleman occupations interacted with economic and political shocks conditional on the measures of preexisting antisemitism and overall occupational segregation interacted with economic and political shocks. Standard errors are corrected for both spatial and temporal correlations following

Table 7, Panel B: Controls for marketwide-shock specification

Spatial matching

fect of the treatment $T1$:	Pogrom occurrence						
mong creditors above the median	(1)	(2)	(3)	(4)	(5)	(6)	(7)
k × Pol. turmoil × 𝟙(Jews among creditors > median)	4.76***	4.30***	4.13***	4.17***	4.07***	4.30***	3.52**
	(1.33)	(1.54)	(1.54)	(1.52)	(1.49)	(1.42)	(1.37)
k × Pol. turmoil × Share of Jews	15.29		16.88	16.93	30.74	17.88	105.31**
	(18.13)		(21.64)	(21.96)	(36.14)	(17.96)	(41.07)
k × Pol. turmoil	1.08	0.70	−0.40	−0.58	−26.24		
	(1.61)	(3.00)	(4.36)	(4.36)	(17.32)		
	0.119	0.152	0.156	0.156	0.162	0.109	0.247
	73,728	24,652	24,652	24,652	24,652	649	800
lent var.	0.522	0.564	0.564	0.564	0.564	3.852	3.250
fect of the treatment $T2$:	Pogrom occurrence						
mong grain traders above the median	(1)	(2)	(3)	(4)	(5)	(6)	(7)
n shock × Pol. turmoil × 𝟙(Jews among gr.traders > median)	1.99**	2.33**	2.18**	2.23**	2.49***	2.23**	2.44**
	(0.93)	(1.02)	(0.99)	(0.98)	(0.96)	(1.06)	(0.98)
n shock × Pol. turmoil × Share of Jews	−2.33		−14.73	−16.62	19.15	−17.53	−5.72
	(10.67)		(22.27)	(26.90)	(29.00)	(32.61)	(23.53)
	0.117	0.108	0.111	0.111	0.125	0.041	0.141
	73,728	27,136	27,136	27,136	27,136	2,120	2,600
lent var.	0.522	0.604	0.604	0.604	0.604	5.802	5.462
ls (M for matched):	All	M	M	M	M	M	M
P for econ & pol. shocks only):	All	All	All	All	All	EP	EP
	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Yes	Yes	Yes	Yes	Yes	No	No
ol pair FE	No	No	No	No	No	No	Yes
h the share of Jews and sectors shares	Yes	No	Yes	Yes	Yes	Yes	Yes
h urbanization rate	No	No	No	Yes	Yes	Yes	Yes
h latitude and longitude	No	No	No	No	Yes	No	No

of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and gri
wise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. This tabl
lts of the spatial matching analysis. The first column presents regressions on the full sample. In columns 2–5, the sample is the matched sample c

5.9 Table 9 and Table 10: Robustness checks

Read:

- Table 9 checks alternative samples, aggregation levels, shock definitions, and political-turmoil definitions.
- Table 10 uses alternative measures of political turmoil and marketwide economic shocks.
- The creditor and grain-trader channels remain robust across these variants.

Economic message: The baseline results do not depend on one narrow definition of shocks, one aggregation level, or one periodization of political turmoil.

TABLE 9

Robustness to changes in the sample, aggregation, and definition of political turmoil, and the construction of climate data

	Pogrom occurrence								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Robustness to:	Climate data:		Definition of political turmoil:			Sample:		Aggregation:	
	without years 1880–82	no data correction	do not add year after	add 2 years after	add 1 year before	post 1861	without large cities	1x1 degree level	uez leve
ɔl turmoil × Sh. Jews among creditors	5.52**	5.58**	6.83**	6.13**	6.78**	5.70**	5.77**	14.08**	9.66
	(2.18)	(2.20)	(3.40)	(2.45)	(3.19)	(2.66)	(2.48)	(6.89)	(3.18)
k × Pol turmoil × Sh. Jews among grain traders	7.65***	7.72***	10.84***	8.37***	7.85***	11.25***	9.07***	21.96***	8.14
	(2.32)	(2.33)	(3.49)	(2.35)	(2.51)	(2.87)	(2.80)	(7.67)	(3.77)
ɔl turmoil	1.00	0.94	5.42***	4.05**	4.87***	4.20***	4.32***	6.01*	6.98
	(0.75)	(0.75)	(2.04)	(1.58)	(1.54)	(1.59)	(1.61)	(3.37)	(3.15)
	72,000	73,728	73,728	73,728	73,728	38,016	72,576	21,376	30,2
	0.117	0.120	0.125	0.122	0.123	0.133	0.119	0.243	0.20
	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ıare of Jews	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ıare of creditors in total employed	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ıare of grain traders in total employed	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
:	0.407	0.522	0.522	0.522	0.522	1.005	0.508	1.244	0.94
	6.366	7.207	7.207	7.207	7.207	9.974	7.112	11.09	9.66

ysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, a
the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. This table explores the robust
sented in column 5 of Table 4. Column 1 drops years 1880–1882 from the sample, column 2 does not apply correction for the local economic sh
nment grain-assistance data, but uses the full sample. Columns 3, 4, and 5 use augmented definition of political turmoil: without a lag added, with
orward added. Column 6 restricts the sample to 1861 and after. Column 7 drops large cities (which served as capitals of any polity in the past or du
Column 8 presents results for the aggregated 1 × 1 degree grid cells. Column 9 replicates the results with the unit of analysis being the district (*ue*
ected for both spatial and temporal correlations following Hsiang (2010) in a radius of 100 km and 1 temporal lag. ***$P < 0.01$, **$P < 0.05$, *$P <$

Table 9: Robustness checks

TABLE 10

Alternative measures of political turmoil and marketwide economic shocks

Panel A: Rising spread as political turmoil; grain price as marketwide shock	Pogrom occurrence		
	(1)	(2)	(3)
Rising sovereign bond yield spread × Local econ shock × Sh. Jews among creditors	5.62*		
	(2.97)		
Rising sovereign bond yield spread × Marketwide econ shock × Sh. Jews among grain traders		6.73*	
		(3.46)	
Log grain price × Political turmoil × Log Jewish grain traders			11.37***
			(3.62)
R^2	0.149	0.148	0.160
Panel B: Baseline specifications on the same samples	**Pogrom occurrence**		
	(1)	(2)	(3)
Political turmoil × Local econ shock × Sh. Jews among creditors	7.10**		
	(3.19)		
Political turmoil × Marketwide econ shock × Sh. Jews among grain traders		8.04*	
		(4.13)	
Marketwide econ shock × Political turmoil × Log Jewish grain traders			2.94***
			(0.79)
R^2	0.151	0.149	0.151
Grid and year FE	Yes	Yes	Yes
Interactions with the share of Jews; all lower-level interactions	Yes	Yes	Yes
Interactions with dummy for former capital cities	Yes	Yes	No
Observations	24,192	24,192	31,225
Mean of dependent var.	1.009	1.009	0.781
SD of dependent var.	9.992	9.992	8.805

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. Panel A presents OLS regressions in which we use alternative measures of political turmoil (columns 1 and 2) and of the marketwide economic shock (column 3). Panel B presents results with the baseline measures on the subsamples where alternative measures are defined. In this table, in addition to the standard set of controls we include a dummy for Kiev, Warsaw, and Vilnius—cities that served as a capital city of any polity in the past or during observation period—interacted with economic and political shocks. This control increases precision of estimates in regressions using the sovereign bond yield spread; and it does not at all affect the baseline estimates. In column 3, instead of the share of Jews among grain traders, we use the log of the number of Jewish grain traders, because this variable has more variation. In Online Appendix Table A12, we show that

Table 10: Alternative shock and turmoil measures

5.10 Table 11 and Table 12: Mechanisms and placebo outcomes

Read:

- Table 11 compares pogrom waves before and after the 1917 state collapse.
- The main effects are present both before and after 1917, suggesting that changes in enforcement capacity alone do not drive the results.
- Table 12 uses placebo crime outcomes such as theft, homicide, and arson.
- The middleman interactions predict pogroms but not general crime outcomes.

Economic message: The mechanism is about ethnic violence against Jewish middlemen rather than a general rise in crime or a simple collapse of state enforcement.

TABLE 11

Enforcement capacity of the state and pogrom determinants

Panel A: The effects of Jewish creditors and local econ shocks	Pogrom occurrence		
	(1)	(2)	(3)
Sample of pogroms waves: 1 (1881–1882); 2 (1903–1906); 3 (1917–1922):	Waves 1 and 2	Wave 3	All waves
Local econ shock × Sh. Jews among creditors	10.62***	6.46*	10.62***
	(3.93)	(3.78)	(3.93)
Local econ shock × Sh. Jews among creditors × $\mathbb{1}$(Wave 3)			−4.16
			(5.45)
Local econ shock × Share of Jews	−6.15	57.19***	−6.15
	(22.20)	(20.49)	(22.20)
Local econ shock × Share of Jews × $\mathbb{1}$(Wave 3)			63.34**
			(30.21)
R^2	0.285	0.279	0.203
Grid FE	Yes	Yes	Yes
Panel B: The effects of Jewish grain traders and marketwide shocks	**Pogrom occurrence**		
	(1)	(2)	(3)
Sample of pogroms waves: 1 (1881–1882); 2 (1903–1906); 3 (1917–1922):	Waves 1 and 2	Wave 3	All waves
Marketwide econ shock × Sh. Jews among grain traders	7.59*	8.57***	7.59*
	(4.23)	(2.85)	(4.23)
Marketwide econ shock × Sh. Jews among grain traders × $\mathbb{1}$(Wave 3)			0.98
			(5.10)
Marketwide econ shock × Share of Jews	−25.76	18.44	−25.76
	(17.41)	(13.95)	(17.41)
Marketwide econ shock × Share of Jews × $\mathbb{1}$(Wave 3)			44.20**
			(22.31)
R^2	0.066	0.043	0.061
Grid FE	No	No	No
Year FE, all lower-level interactions	Yes	Yes	Yes
Observations	3,456	3,456	6,912
Mean of dependent var.	6.887	3.993	5.440
SD of dependent var.	25.33	19.58	22.68

Notes: The unit of analysis is grid cell × year. The dependent variable is a dummy variable that takes the value of 1 if a pogrom occurred in a given year and grid cell, and 0 otherwise. We multiply the dependent variable by 100 to measure all coefficients in percentage points of probability of pogrom occurrence. The sample in column 1 includes only years of the first two waves of pogroms. The sample in column 2 includes only years of the third wave of pogroms. The sample of column 3 includes years of all pogrom waves. Standard errors are corrected for both spatial and temporal correlations following Hsiang (2010) in a radius of 100 km and 1 temporal lag. ***$P<0.01$, **$P<0.05$, *$P<0.1$

Table 11: State capacity and pogrom determinants

TABLE 12

Placebo: general crime, 1900–1912

Panel A: The effects of local econ shocks	Pogrom occurrence	Theft per capita	Homicide per capita	Arson per capita
	(1)	(2)	(3)	(4)
Local econ shock × Political turmoil × Sh. Jews among creditors	9.84***	−98.87	−8.91	−77.83
	(3.44)	(76.71)	(9.13)	(59.32)
Local econ shock × Political turmoil × Share of Jews	−6.11	413.97	35.03	298.28
	(20.33)	(334.49)	(45.93)	(254.76)
R^2	0.205	0.921	0.831	0.915
Panel B: The effects of marketwide econ shocks	**Pogrom occurrence**	**Theft per capita**	**Homicide per capita**	**Arson per capita**
	(1)	(2)	(3)	(4)
Marketwide econ shock × Political turmoil × Sh. Jews among grain traders	3.77*	91.60	30.78	7.81
	(2.13)	(88.67)	(23.79)	(13.70)
Marketwide econ shock × Political turmoil × Share of Jews	14.79	−168.53	−40.87	−11.75
	(9.43)	(145.73)	(31.98)	(9.54)
R^2	0.201	0.921	0.831	0.915
Grid and year FE	Yes	Yes	Yes	Yes
Interactions with the share of Jews; all lower-level interactions	Yes	Yes	Yes	Yes
Observations	7,488	6,422	6,422	5,434
Mean of dependent var.	1.950	50.51	7.758	20.45
SD of dependent var.	13.83	624.6	102.9	287.7

Notes: The unit of analysis is grid cell × year. This table considers theft, homicide, and arson per capita as placebo outcomes, and reports results for pogrom occurrence for the sample of years in which the placebo outcomes are available: 1900–1912. Data on arson have even shorter time span: 1900–1910. Due to the smaller sample, we make a less conservative assumption about clusters in this table than in other tables: standard errors are clustered at grid-cell level without accounting for spatial correlation. ***$P<0.01$, **$P<0.05$, *$P<0.1$

Table 12: Placebo general crime outcomes

6 Short summary

- The paper studies anti-Jewish pogroms in the Pale of Settlement from 1800 to 1927.
- Pogroms clustered in waves that occurred when economic shocks coincided with political turmoil.
- Jewish occupational specialization matters more than the general Jewish population share.
- Local crop shocks during political turmoil increased pogroms in places where Jews dominated moneylending.
- Marketwide grain shocks during political turmoil increased pogroms in places where Jews dominated grain trading.
- Other Jewish occupations do not show the same robust positive effect.
- Pre-existing antisemitism, occupational segregation, and spatial differences do not explain away the results.
- Spatial matching and many robustness checks support the main findings.
- Placebo outcomes show that the mechanism is specific to pogroms rather than general crime.
- The interpretation is that repeated middleman relationships provided informal insurance in stable times, but broke down during political turmoil.

7 Conclusion

7.1 Core conclusion

Grosfeld, Sakalli, and Zhuravskaya show that anti-Jewish pogroms in the Russian Empire occurred when economic shocks and political turmoil coincided, especially in places where Jews dominated agriculturally related middleman occupations. The pattern is not consistent with indiscriminate scapegoating. It is consistent with a breakdown of economic relationships between majority peasants or grain buyers and Jewish creditors or grain traders.

7.2 Economic intuition

Middlemen can be valuable because they provide credit, trade links, and insurance. In ordinary times, repeated interaction supports cooperation: creditors can roll over debt, and traders can maintain relationships. Political turmoil changes incentives. When the future is uncertain and enforcement weakens, middlemen demand immediate repayment or stop providing insurance. Debtors and buyers then have stronger incentives to attack the middleman minority, producing violence.

7.3 Takeaway

The paper gives a politico-economic explanation for violence against middleman minorities. Middleman minorities may be vulnerable not because they compete with the majority, but because they occupy key nodes in economic relationships that become fragile during crises. The main lesson is that ethnic violence can arise from the collapse of complementary economic relationships under political uncertainty.